SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1708

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour
QUESTIONS: 3 Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I S.Sanjai(192421121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.SANJAI

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
         A
        / \
       B   C
      /\   /\
     D  E F  G
```

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
     A

    / \

   B   C

  /\   /\

 D  E F  G
```

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**

```
1 3 6
5 0 2
4 7 8
```

Goal State :

1	2	3
4	5	6
7	8	0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.
Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Breadth-First Search (BFS)

```
Iteration        Queue            Visited Node
1                ['A']    A
2                ['B', 'C']       B
3                ['C', 'D', 'E'] C
4                ['D', 'E', 'F', 'G']     D
5                ['E', 'F', 'G'] E
6                ['F', 'G']       F
7                ['G']    G

BFS Traversal Order:
A -> B -> C -> D -> E -> F -> G
```

Depth-First Search (DFS).

```
Iteration        Stack            Visited Node
1                ['A']    A
2                ['C', 'B']       B
3                ['C', 'E', 'D'] D
4                ['C', 'E']       E
5                ['C']    C
6                ['G', 'F']       F
7                ['G']    G

DFS Traversal Order:
A -> B -> D -> E -> C -> F -> G
```

3 – 8 Puzzle using Breadth-First Search (BFS).

```
(5, 2, 0)
(4, 7, 8)

Step 2
(1, 3, 0)
(5, 2, 6)
(4, 7, 8)

Step 3
(1, 0, 3)
(5, 2, 6)
(4, 7, 8)

Step 4
(1, 2, 3)
(5, 0, 6)
(4, 7, 8)

Step 5
(1, 2, 3)
(0, 5, 6)
(4, 7, 8)

Step 6
(1, 2, 3)
(4, 5, 6)
(0, 7, 8)

Step 7
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 8
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)

Total States Generated: 456
Number of Moves: 8
```